Unveiling the ***Environmental Cost of Fast Fashion***

A Data Engineering End-to-End Analysis

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Role*: Junior Data Engineer*

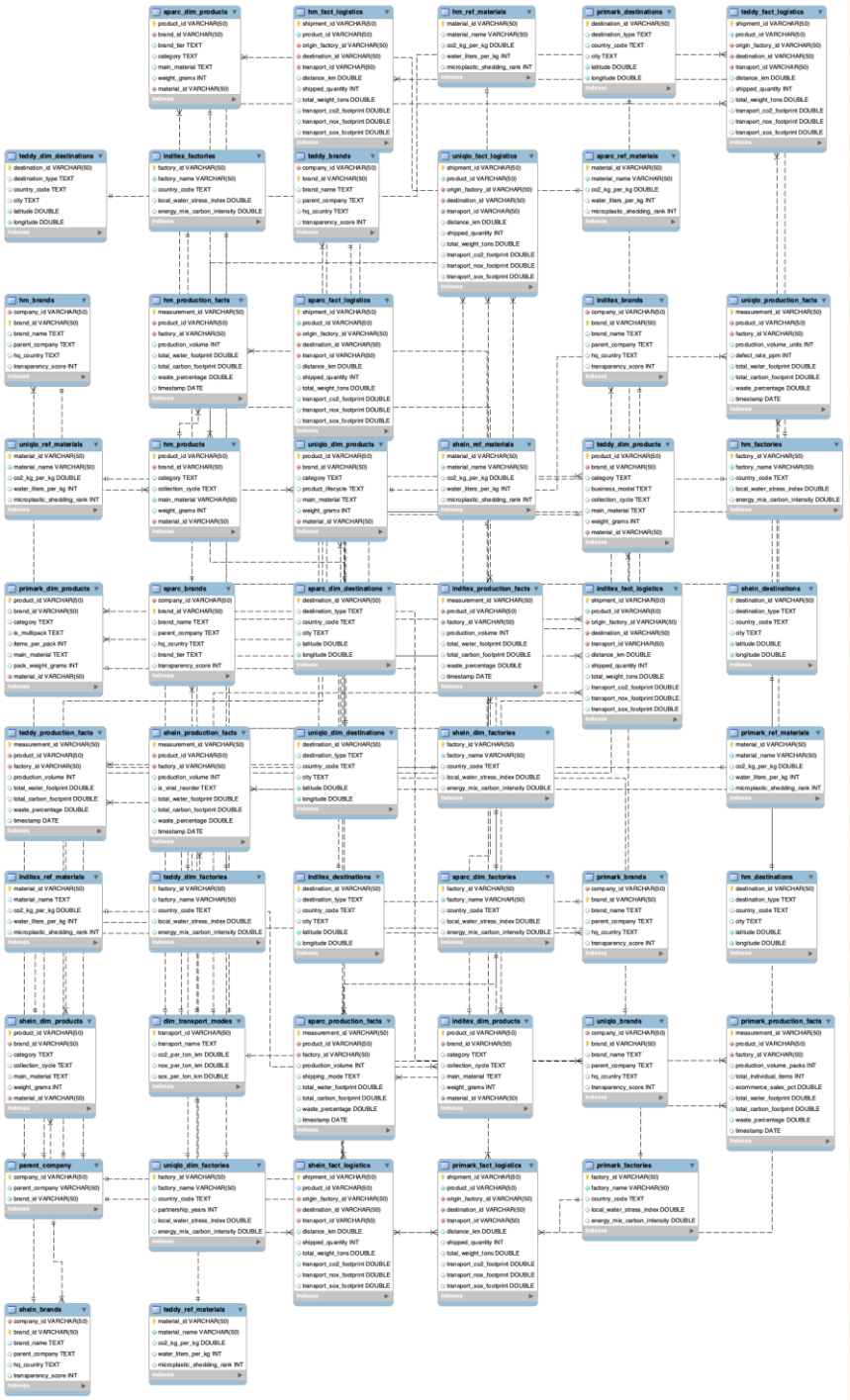
Objective: Quantifying the ecological footprint of 7 global brands (H&M, Inditex, Uniqlo, Primark, Shein, Sparc, Teddy)

- **The Pipeline:**
- **Data Ingestion:** Cleaning raw CSVs via PowerQuery (Excel).
- **Modeling:** Designing a relational Star Schema in MySQL.
- **ETL & Logic:** Modular SQL scripts (Functions, Procedures, Views).
- **Visualization:** Interactive dashboards in Tableau and Python (Seaborn/Matplotlib).

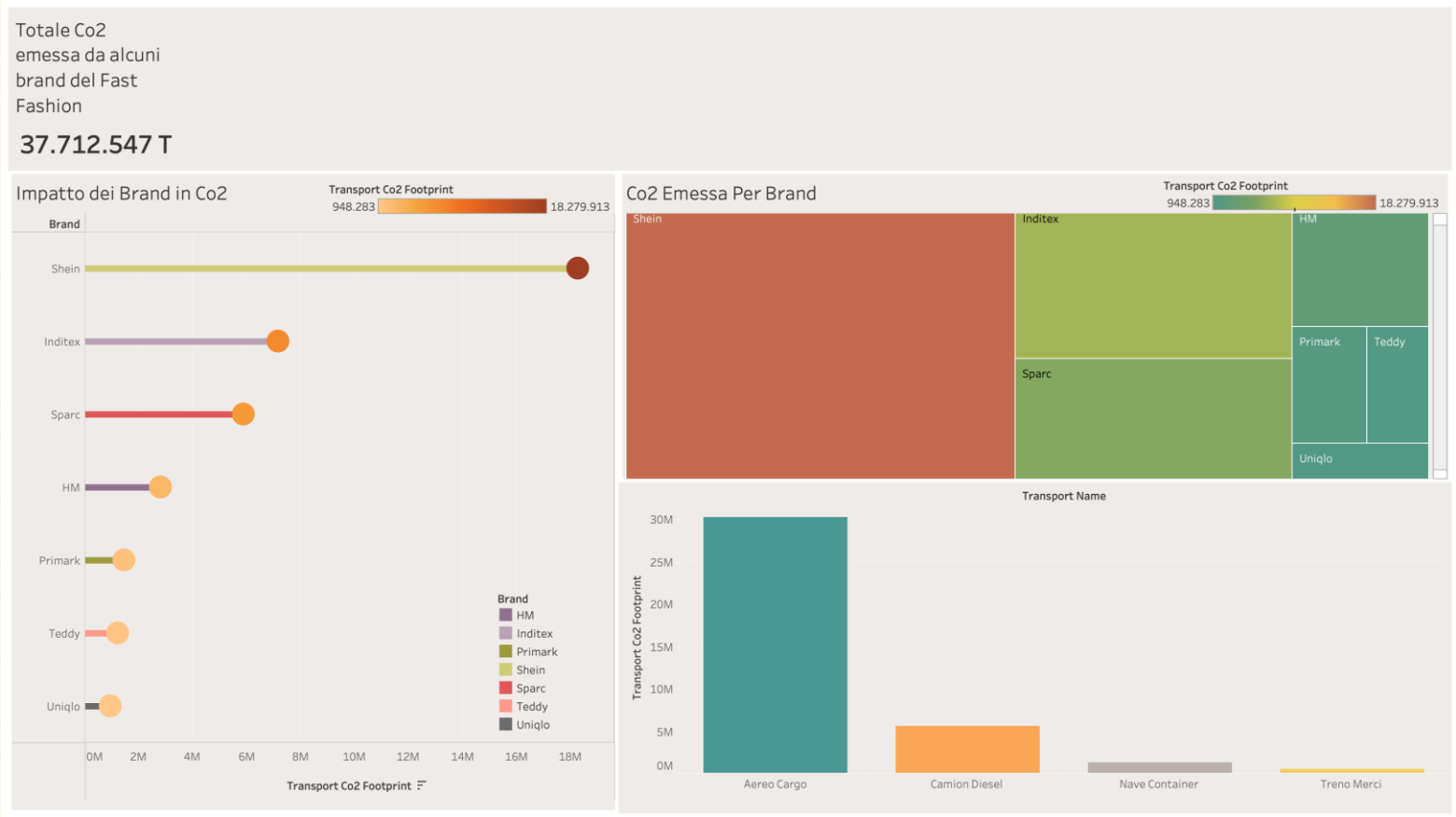
Data Modeling
(*ER Schema*)

Core Logic: Designed a scalable architecture to link production volumes, logistics facts, and raw material properties.

Technical Highlight: Implementation of a Star Schema to optimize analytical performance and ensure data integrity across 7 different brands.



Carbon Footprint & Logistics Insights



SQL Logic: Used custom functions (CalcolaCarbonFootprintProdotto) to calculate total emissions by merging production and logistics data.

Key Finding: Air Cargo represents the highest carbon-intensive transport mode, significantly impacting the overall brand footprint.

Total Impact: Analyzed a total of 37.7 Million Tons of CO2 across the sector.

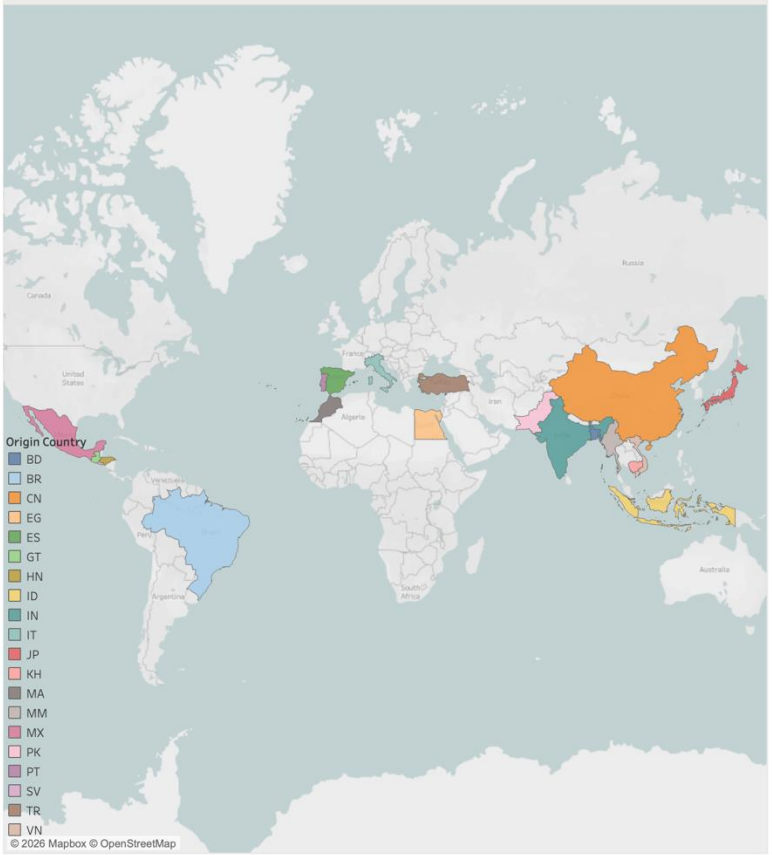
Water & Microplastics Analysis

SQL Logic: Leveraged joins between production facts and material reference tables to calculate weighted averages of water consumption.

Key Finding: High reliance on synthetic materials leads to a combined microplastic release of 435.6 mg/kg after the first wash.

Efficiency: Identified specific material-brand combinations that exceed 12,000 liters of water per garment.

ESG Ranking & Global Mapping



SQL Logic: Engineered a multi-variable scoring algorithm using CASE WHEN logic to assign monthly sustainability grades (A-F).

Geospatial Data: Mapped production hotspots using geospatial coordinates from dim_factories.

Trend: Performance ratings fluctuate significantly, showing a direct correlation between material choice and the final ESG score.

Technical Conclusions & Scalability

- **Automation:** The SQL-based architecture allows for automated report updates by simply importing new monthly data into the raw layer.
- **Modularity:** Modular scripts (01-04) ensure the system can easily integrate new brands or environmental metrics.
- **Future Work:** Potential for real-time API integration and predictive modeling to forecast future environmental risks.

Contact & Repository

Thank You!

GitHub Link: https://github.com/ElenaPiracci/fast-fashion-data-engineering/tree/main

LinkedIn Profile: www.linkedin.com/in/elenapiracci